Zotero Tag Reference Sheet

ORISE Infographic Submission (Graphite as a Critical Material)

Color Assignments

Table 1. The outline of a high-visibility color-coding scheme for Zotero tags as well as the reasoning behind it.

Color	Tag	Reason
Green	`info-datapoint`	To spot sources with usable statistics
Blue	`paper-supports-claim`	Core evidence for argument
Red	`sq-needs-verification`	Sources to be flagged for follow-up
Orange	`paper-contradicts-claim`	Counterevidence against argument
Purple	`info-map-data`	Geographic data for map visuals
Yellow	`sq-high-priority`	Must-read or must-cite sources

Topical Tags

Graphite Fundamentals (Prefix: `gf-`)

Table 2. Zotero tagging system for the fundamental information with regards to graphite.

Tag	Reason
`gf-properties`	Physical and chemical properties (conductivity, lubricity, etc.)
`gf-types`	Distinction between natural flake, amorphous, vein/lump, etc.
`gf-geology`	Formation, ore deposits, mineralogy, etc.
`gf-processing`	Beneficiation, purification, spheronization, coating, etc.

Supply Chain and Geopolitics (Prefix: `sc-`)

Table 3. Zotero tagging system for the supply chain and geopolitical information of graphite.

Tag	Reason
`sc-mining-global`	Global mining operations and overview of industry
`sc-reserves`	Probable reserves and resource estimates by country
`sc-production-china`	China-specific production and dominance
`sc-production-africa`	Mozambique, Madagascar, or other African producers
`sc-production-na`	US or Canada mining projects or production data
`sc-trade-policy`	Tariffs, trade agreements, and import/export data flows
`sc-export-restrictions`	Export bans, quota systems, etc.
`sc-risk`	Supply chain vulnerabilities and disruption scenarios
`sc-concentration`	Market concentration metrics and HHI indices
`sc-foreign-dependency`	A nation's reliance on foreign graphite sources

Critical Mineral Designation and Policy (Prefix: `pol-`)

Table 4. Zotero tagging system for what defines a "critical mineral" and policies regarding it.

Tag	Reason
`pol-cm-definition`	What "critical mineral" means
`pol-cm-criteria`	Specific scoring or assessment frameworks
`pol-cm-list-US`	USGS critical minerals list and US-specific designations
`pol-cm-list-EU`	EU Critical Raw Materials Act list and assessments
`pol-cm-list-other`	Australia, Canada, Japan, or other critical minerals list
`pol-IRA`	Inflation Reduction Act provisions about critical minerals
`pol-DoD`	Department of Defense stockpiling reserves
`pol-DPA`	Defense Production Act invocations for supply chains
`pol-EU-CRM-act`	European Critical Raw Materials details
`pol-rsrc-nationalism`	Government control of mineral sources

Applications and Demands (Prefix: `app-`)

Table 5. Zotero tagging system for specific application cases of graphite.

Tag	Reason
`app-batteries`	Lithium-ion battery anodes or graphite specs in batteries
`app-EV`	Electric vehicle market
`app-steel`	Refractory use, crucibles, and steelmaking
`app-lubricants`	Application of graphite in lubricant and grease
`app-nuclear`	Nuclear energy applications
`app-electronics`	Electronic uses, thermal management, etc.
`app-demand-forecast`	Projected future demand and growth models
`app-demand-drivers`	Factors increasing demand (EV adoption, energy, etc.)
`app-market-size`	Total addressable market, market valuation, etc.
`app-price-trends`	Historical and projected pricing, cost per ton, etc.

Environment and Sustainability (Prefix: `env-`)

Table 6. Zotero tagging system for the impact of graphite mining on the environment and such.

Tag	Reason
`env-impact-mining`	Environmental damage as a result of graphite extraction
`env-lifecycle`	Lifecycle assessment and cradle-to-grave analysis
`env-carbon`	Carbon footprint of graphite production and use
`env-tailings`	Tailings, waste management, and acid mine drainage
`env-water`	Water consumption or contamination from processing
`env-land`	Land use, reclamation, habitat disruption
`env-ESG`	ESG standards and sourcing frameworks
`env-community`	Community impact, labor impacts, and displacement

Alternatives, Recycling, and Innovation (Prefix: `alt-`)

Table 7. Zotero tagging system for possible alternatives for graphite and other novelties.

Tag	Reason
`alt-recycling`	Graphite recovery from spent batteries
`alt-urban-mining`	Recovery of graphite from waste streams or used products
`alt-synthetic`	Synthetic graphite production or petroleum coke feedstock
`alt-substitutes`	Silicon anodes, hard carbon, and other anode materials
`alt-graphene`	Graphene as a derivative or related advanced material
`alt-R&D`	Processing tech advancements and purification methods
`alt-circular-economy`	Circular supply chain models and closed-loop systems

Economic and Market Structure (Prefix: `econ-`)

Table 8 Zotero tagging system for the graphite market and the economy as a result.

Tag	Reason
`econ-market-players`	Major companies within the graphite market
`econ-investment`	Investment trends and government subsidies
`econ-junior-mining`	Junior mining companies and exploration-stage projects
`econ-value-chain`	Downstream processing, etc.
`econ-cost-production`	Cost breakdowns and operating costs
`econ-impact-local`	Economic impact on mining communities and job creation